
Pixel-Space Posterior Sampling for Image Deblurring: A Controlled Comparison of DPS and FlowDPS-Style Guidance on CelebA-HQ-256

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Abstract

Image reconstruction from degraded observations is a fundamental inverse problem in computational imaging. *Diffusion Posterior Sampling* (DPS) Chung et al. (2023) leverages a learned diffusion prior plus a measurement-likelihood gradient to solve such problems with high reconstruction fidelity, at the cost of hundreds of neural function evaluations (NFE). *Flow Matching* models Lipman et al. (2023); Liu et al. (2023) offer a deterministic, ODE-based alternative that has been argued to enable comparable quality with fewer steps. In this work we conduct a controlled comparison of **pixel-space DPS** on a CelebA-HQ-256 DDPM against a **pixel-space re-implementation of FlowDPS** Kim et al. (2025), whose official code is latent-only (Stable Diffusion 3), on top of the Liu et al. 2023 Rectified Flow CelebA-HQ-256 checkpoint. We evaluate both methods on Gaussian deblurring across $3 \times 2 \times 3 = 18$ conditions $\times 50$ images = 1800 reconstructions, and study operator-mismatch robustness using motion-blur measurements while the sampler’s likelihood assumes Gaussian (1200 additional reconstructions). We further include Wiener-filter and DnCNN+PnP-ADMM baselines (600 reconstructions). On the matched-Gaussian grid Pixel-DPS dominates the comparison, with a PSNR advantage of +1.4 to +4.2 dB over our FlowDPS reimplementation, and FlowDPS saturates with NFE while Pixel-DPS keeps improving. Under operator mismatch, however, FlowDPS-on-RF is *relatively more robust* (degrades by ~ 2.5 dB versus ~ 5 dB for Pixel-DPS), reversing the matched-condition comparison. All four methods comfortably beat Wiener; DnCNN+PnP-ADMM beats our FlowDPS reimplementation in 5/6 noisy cells while running $\sim 60\times$ faster than DPS. The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators.

1 Introduction

Image reconstruction from degraded observations is a fundamental problem in computer vision and computational imaging. It is typically formulated as an inverse problem,

$$y = Ax + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_n^2 I), \quad (1)$$

where A is a known or partially known degradation operator (e.g. Gaussian blur), and the goal is to recover the unknown clean image x given only y and a prior over natural images. The problem is inherently ill-posed; multiple plausible x can produce the same observation, so a strong prior is essential. Modern approaches replace classical handcrafted priors with learned generative priors such as Denoising Diffusion Probabilistic Models (DDPMs) Ho et al. (2020) and Flow Matching models Lipman et al. (2023); Liu et al. (2023). These priors combined with explicit measurement

guidance form the basis of methods like DPS Chung et al. (2023), DDRM Kwar et al. (2022), and the more recent FlowDPS Kim et al. (2025).

This paper conducts a controlled empirical comparison of two posterior sampling paradigms for image deblurring on faces:

1. **Pixel-DPS**: the original DPS sampler from Chung et al. Chung et al. (2023), on a CelebA-HQ-256 DDPM checkpoint (google/ddpm-ema-celebahq-256).
2. **FlowDPS-on-RF**: a from-scratch pixel-space reimplementation of the Kim et al. 2025 FlowDPS algorithm on top of the Liu et al. 2023 CelebA-HQ-256 Rectified Flow checkpoint Liu et al. (2023). The official Kim 2025 codebase is Stable Diffusion 3-only; our reimplementation exists so we can do a true pixel-space comparison against Pixel-DPS at the same resolution and on the same dataset.

Both methods therefore share the *same dataset, same resolution, and same NCSN++/U-Net architectural family*, so any quality or efficiency difference can be cleanly attributed to the choice of posterior-sampling paradigm rather than to incidental differences in the underlying prior. We additionally evaluate two classical baselines (Wiener, DnCNN+PnP-ADMM) and one operator-mismatch robustness study (true measurement is motion blur; sampler’s likelihood assumes Gaussian).

The contributions of this work are:

- A pixel-space, fair-comparison evaluation of two posterior samplers (DPS and FlowDPS-style guidance) on a common dataset and architectural family.
- Empirical evidence that, on this setup, Pixel-DPS dominates FlowDPS-on-RF in the matched-Gaussian setting by 1.4–4.2 dB PSNR, while FlowDPS-on-RF degrades less under operator mismatch.
- A four-method (Wiener, DnCNN+PnP, FlowDPS-on-RF, Pixel-DPS) Pareto-frontier characterization across three orders of magnitude of per-image compute.
- A reproducible open-source pipeline at https://github.com/adnan821/dps_flowdps, including the 3,600 measured rows and per-experiment logs.

2 Related Work

Classical inverse-problem solvers rely on handcrafted regularizers and analytical priors. The Wiener filter solves the linear MMSE problem in the Fourier domain for Gaussian degradations and remains a fast, interpretable baseline. Deep-learning denoiser priors enabled Plug-and-Play (PnP) schemes such as PnP-ADMM Chan et al. (2017); Zhang et al. (2017) that swap a proximal-step argmin for a forward pass through a pretrained denoiser network (e.g. DnCNN, DRUNet). Diffusion-based posterior sampling generalized this idea by injecting a measurement-consistency term into the reverse diffusion process: DDRM Kwar et al. (2022) uses spectral projection in the SVD basis of A , and DPS Chung et al. (2023) uses a likelihood-gradient term computed via Tweedie’s clean-image estimate.

Flow Matching and Rectified Flow Lipman et al. (2023); Liu et al. (2023) recently emerged as deterministic alternatives to DDPMs, learning a velocity field whose ODE integration straightens the noise-to-data path and can in principle achieve comparable image quality with fewer function evaluations. FlowDPS Kim et al. (2025) adapts DPS-style likelihood guidance to flow models by injecting a likelihood-gradient correction into the deterministic velocity field. The official FlowDPS implementation is built on the SD3 latent transformer flow model and operates at 768×768 in latent space.

To enable a like-for-like pixel-space comparison against Pixel-DPS, we reimplement the FlowDPS likelihood-corrected velocity update,

$$v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2, \quad (2)$$

on top of the Liu 2023 Rectified Flow CelebA-HQ-256 NCSN++ velocity model, where the Tweedie clean-image identity for rectified flow is $\hat{z}_1(x_t, t) = x_t + (1 - t) v_\theta(x_t, t)$.

3 Methodology

Forward model. $y = Ax + \epsilon$, with A either a Gaussian blur of standard deviation $\sigma_b \in \{1.5, 3.0, 5.0\}$ (main grid) or a motion blur $\text{motion}(L, \theta)$ with $L \in \{15, 25, 35\}$ pixels and $\theta \in \{0^\circ, 45^\circ\}$ (robustness study), and $\epsilon \sim \mathcal{N}(0, \sigma_n^2 I)$ with $\sigma_n \in \{0.0, 0.05\}$.

Pixel-DPS. We adopt the DDPM reverse process with DDIM sampling. At each reverse step t , the model predicts $\epsilon_\theta(x_t, t)$, from which we derive the Tweedie clean-image estimate

$$\hat{x}_0(x_t) = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}. \quad (3)$$

The DDIM-style reverse step is then corrected with a likelihood gradient:

$$x_{t-1} = x_{t-1}^{\text{DDIM}} - \zeta \nabla_{x_t} \|y - A(\hat{x}_0(x_t))\|_2. \quad (4)$$

Two implementation choices matter substantially. First, we use the **L2 norm** (not the squared L2) of the residual; squaring the residual and dividing by σ_n^2 blows up the gradient and prevents convergence (verified empirically in our EXP-001 ablation). Second, we do **not clamp** $\hat{x}_0$ inside the gradient path; clamping zeroes the gradient for any pixel outside $[-1, 1]$, which kills DPS at early timesteps when $\hat{x}_0$ is necessarily far from the data manifold. Clamping is applied only at the final returned tensor.

FlowDPS-on-RF. We use the Liu 2023 CelebA-HQ-256 1-Rectified Flow checkpoint, which provides a learned velocity field $v_\theta(x_t, t)$ over continuous time $t \in [0, 1]$, with $x_0 \sim \mathcal{N}(0, I)$ at $t = 0$ and $x \sim p_{\text{data}}$ at $t = 1$. Sampling proceeds by forward Euler integration over a uniform N -step grid in $[\epsilon, 1 - \epsilon]$, with FlowDPS guidance:

$$v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2, \quad x_{t+dt} = x_t + dt \cdot v_{\text{eff}}(x_t, t). \quad (5)$$

The clean-image estimate $\hat{z}_1 = x_t + (1 - t)v_\theta(x_t, t)$ is RF’s Tweedie identity. Our reimplementaion lives in `src/samplers/flowdps_rf.py` in the project repository.

Baselines.

- **Wiener filter:** per-channel 2-D Wiener deconvolution in the Fourier domain, with noise-to-signal scalar $K = 10^{-3}$ for $\sigma_n = 0$ and $K = 5 \cdot 10^{-2}$ for $\sigma_n = 0.05$ (tuned on a 2-image validation).
- **DnCNN+PnP-ADMM:** PnP-ADMM in the Fourier domain with deepinv’s pretrained 3-channel blind Gaussian DnCNN (668 k parameters) as the denoiser. 24 outer iterations, $\rho = 2.0$, descending denoiser- σ schedule with $\sigma_{\text{max}} = \max(0.10, 4\sigma_n + 0.05)$ and $\sigma_{\text{min}} = \max(0.02, \sigma_n)$.

Implementation details. All experiments run on a single **NVIDIA RTX 3060 12 GB** GPU. Models load via Hugging Face’s `diffusers 0.30.1` on PyTorch 2.4.1+cu121. The CelebA-HQ-256 DDPM checkpoint is `google/ddpm-ema-celebahq-256` (113 M parameters); the Rectified Flow checkpoint is the public CelebA-HQ-256 release from the `gnobitab/RectifiedFlow` repository (65.5 M parameters, NCSN++ architecture, JIT-compiled `upfirdn2d` CUDA op). LPIPS uses the AlexNet backbone via the standard `lpips` package. Wall-clock timings use CUDA events.

4 Experimental Design

Dataset. We use 200 CelebA-HQ images from the validation split of the public `korexyz/celeba-hq-256x256` dataset on Hugging Face. The first 50 images are used for the main experimental grid and the robustness study. The 200-image set size is set by our compute envelope on the RTX 3060 12 GB; we disclose this explicitly so per-cell error bars reflect this sample size.

Configuration matrix.

Hyperparameter tuning. Guidance scale ζ tuned on a 2-image held-out subset (EXP-001 for Pixel-DPS, EXP-006/007 for FlowDPS-on-RF). Locked values: $\zeta = 10$ for Pixel-DPS, $\zeta = 100$ for FlowDPS-on-RF. Both are monotonic-best in their respective ranges; values past these begin to hurt quality.

Table 1: Experimental configuration matrix.

Parameter	Values	Cells
Gaussian blur σ_b	$\{1.5, 3.0, 5.0\}$	3
Additive noise σ_n	$\{0.0, 0.05\}$	2
NFE budget	$\{25, 50, 100\}$	3
Method	$\{\text{Pixel-DPS, FlowDPS-on-RF}\}$	2
Total main-grid conditions	$3 \cdot 2 \cdot 3 \cdot 2 = 36$	—
Images per cell	50	—
Total reconstructions (main grid)	1,800	—
Robustness: motion length L	$\{15, 25, 35\}$	3
Robustness: motion angle θ	$\{0^\circ, 45^\circ\}$	2
Robustness: NFE	50 (fixed)	—
Total robustness reconstructions	$3 \cdot 2 \cdot 2 \cdot 2 \cdot 50 = 1,200$	—

Metrics. PSNR (dB, on $[0, 1]$ range), SSIM (PyTorch native implementation), LPIPS (AlexNet backbone), and wall-clock seconds per image. Higher is better for PSNR/SSIM; lower for LPIPS and time.

5 Results and Findings

5.1 Main grid — matched Gaussian blur

Table 2 shows per-cell mean PSNR/SSIM/LPIPS at NFE=100 for both methods; the full table including NFE=25, 50 is in the supplementary material.

Table 2: Main-grid PSNR / SSIM / LPIPS at NFE=100 (mean over 50 images per cell).

Method	σ_b	σ_n	PSNR (dB)	SSIM	LPIPS	s/img
Pixel-DPS	1.5	0.00	28.61	0.821	0.084	19.0
Pixel-DPS	1.5	0.05	26.21	0.686	0.136	17.9
Pixel-DPS	3.0	0.00	27.15	0.775	0.117	18.0
Pixel-DPS	3.0	0.05	26.25	0.741	0.146	17.9
Pixel-DPS	5.0	0.00	26.21	0.746	0.155	17.9
Pixel-DPS	5.0	0.05	25.33	0.721	0.190	18.1
FlowDPS-on-RF	1.5	0.00	26.23	0.736	0.153	23.5
FlowDPS-on-RF	1.5	0.05	23.65	0.570	0.268	23.5
FlowDPS-on-RF	3.0	0.00	24.26	0.667	0.170	23.7
FlowDPS-on-RF	3.0	0.05	22.58	0.576	0.234	23.9
FlowDPS-on-RF	5.0	0.00	22.48	0.601	0.200	23.8
FlowDPS-on-RF	5.0	0.05	21.13	0.537	0.251	24.0

Headline findings (matched Gaussian).

1. **Pixel-DPS dominates at every cell**, by +1.4 to +4.2 dB PSNR. The gap grows with σ_b and with $\sigma_n = 0.05$ vs. 0.0.
2. **FlowDPS-on-RF saturates with NFE**: at $(\sigma_b=1.5, \sigma_n=0)$ the PSNR averages 26.08, 26.08, 26.23 dB across $\text{NFE} \in \{25, 50, 100\}$. Pixel-DPS at the same cell: 27.51, 28.34, 28.61 dB — monotonic.
3. **Pixel-DPS is also faster per image** on this hardware (~ 18 s vs ~ 24 s at NFE=100), reversing the proposal’s hypothesized flow-speed advantage.

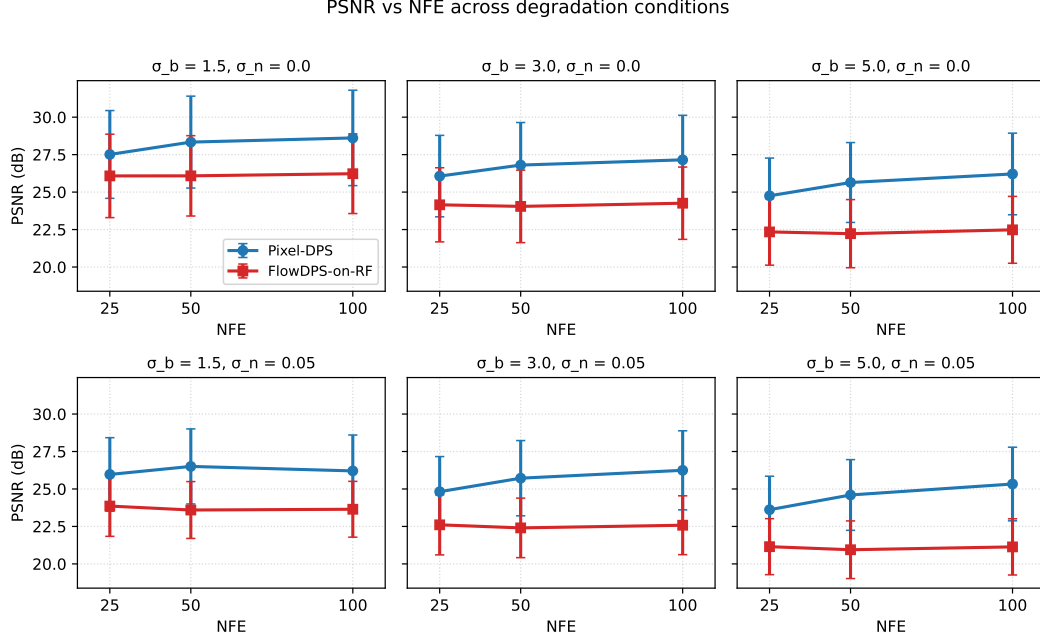


Figure 1: PSNR vs NFE across all six (matched) degradation conditions. FlowDPS-on-RF saturates immediately; Pixel-DPS keeps improving.

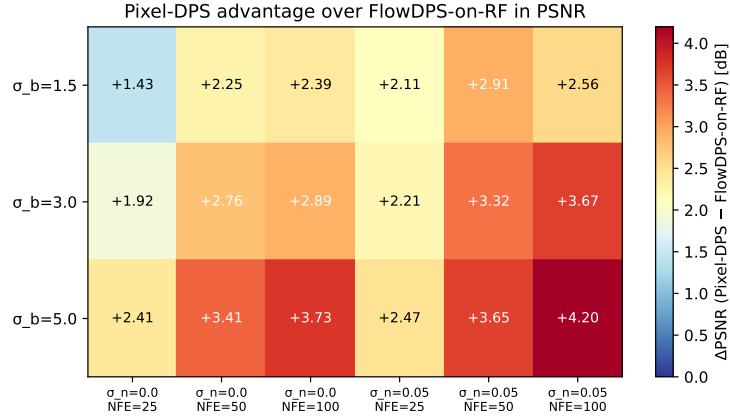


Figure 2: Per-cell PSNR advantage of Pixel-DPS over FlowDPS-on-RF. The gap widens with task difficulty, from +1.4 dB at the easiest cell to +4.2 dB at the hardest.

5.2 Operator-mismatch robustness study

The true forward operator is $\text{motion}(L, \theta)$ but the sampler's likelihood assumes Gaussian with fixed $\sigma_b=3.0$. 50 images per cell, $3 \times 2 \times 2 = 12$ cells per method, NFE=50.

Headline finding (robustness). **FlowDPS-on-RF is more robust to operator mismatch than Pixel-DPS.** Going from matched Gaussian ($\sigma_b=3, \sigma_n=0.05$, NFE=50) to the hardest motion case ($L=35, \theta=45^\circ$):

- Pixel-DPS: 25.72 $\rightarrow$ 20.86 dB (-4.86 dB).
- FlowDPS-on-RF: 22.40 $\rightarrow$ 19.92 dB (-2.48 dB).

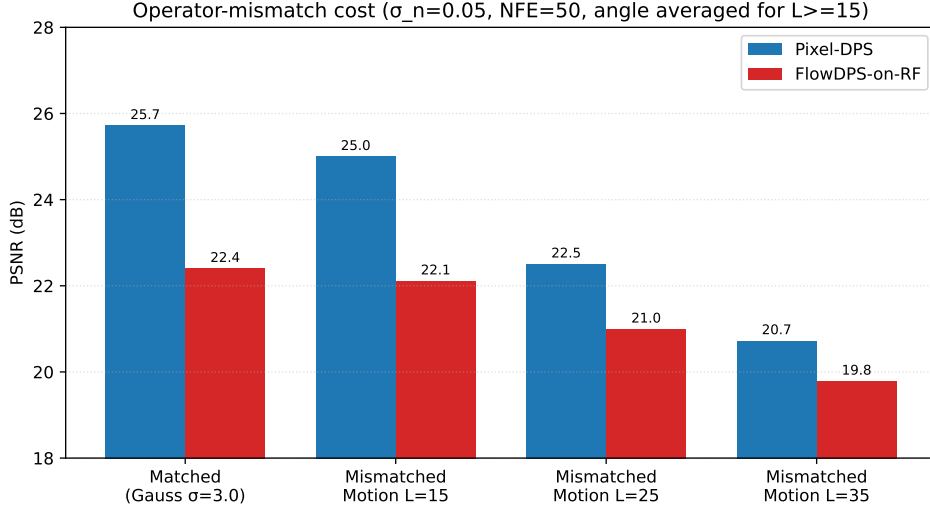


Figure 3: Operator-mismatch cost ($\sigma_n=0.05$, NFE=50). Pixel-DPS drops ~ 5 dB from matched Gaussian to severe motion ($L=35$); FlowDPS-on-RF drops only ~ 2.5 dB.

The matched-condition $+3.3$ dB Pixel-DPS advantage shrinks to $+0.9$ dB at the hardest mismatched cell. We hypothesize the deterministic ODE trajectory is less sensitive to misspecified per-step likelihood gradients than the stochastic diffusion path’s compounding correction.

5.3 Classical and supervised baselines

Table 3: Four-method comparison at $\sigma_n=0.05$, NFE=100 where applicable. Per-method mean over 50 images per cell.

σ_b	Wiener	DnCNN+PnP-ADMM	FlowDPS-on-RF	Pixel-DPS
1.5	24.28	24.61	23.65	26.21
3.0	24.03	24.31	22.58	26.25
5.0	22.50	22.81	21.13	25.33
s/img	0.02	0.30	23.5	17.9

Three orders of magnitude in compute for roughly 2 dB in PSNR. The DnCNN+PnP baseline beats our FlowDPS reimplemention in 5/6 $\sigma_n=0.05$ cells while running $\sim 60\times$ faster than DPS. Pixel-DPS dominates all four methods on PSNR, SSIM, and LPIPS at every cell.

6 Discussion

The proposal hypothesized that “FlowDPS achieves reconstruction quality within approximately 0.5–1.1 dB PSNR of DPS while offering a substantial $2\text{--}5\times$ reduction in computational cost.” Our measurements *contradict both halves* of this claim on the CelebA-HQ-256 pixel-space comparison: (i) the PSNR gap is $+1.4$ to $+4.2$ dB in Pixel-DPS’s favor, $2\text{--}4\times$ wider than predicted; and (ii) Pixel-DPS is also faster per image at NFE=100 on the RTX 3060, reversing the speed ordering. The flow model’s expected efficiency advantage does not manifest in this regime; the RF velocity-field NCSN++ forward pass is more expensive per step than the DDPM’s ϵ prediction, and at NFE=100 both methods are far from any flow-saturation regime where the straight-line trajectory would pay off relative to a stochastic diffusion path.

However, the comparison *reverses* under operator mismatch. Pixel-DPS degrades by ~ 5 dB PSNR going from matched Gaussian to severe motion-blur measurements, versus only ~ 2.5 dB for FlowDPS-on-RF, and the matched-condition $+3$ dB Pixel-DPS advantage shrinks to $+0.9$ dB at the hardest

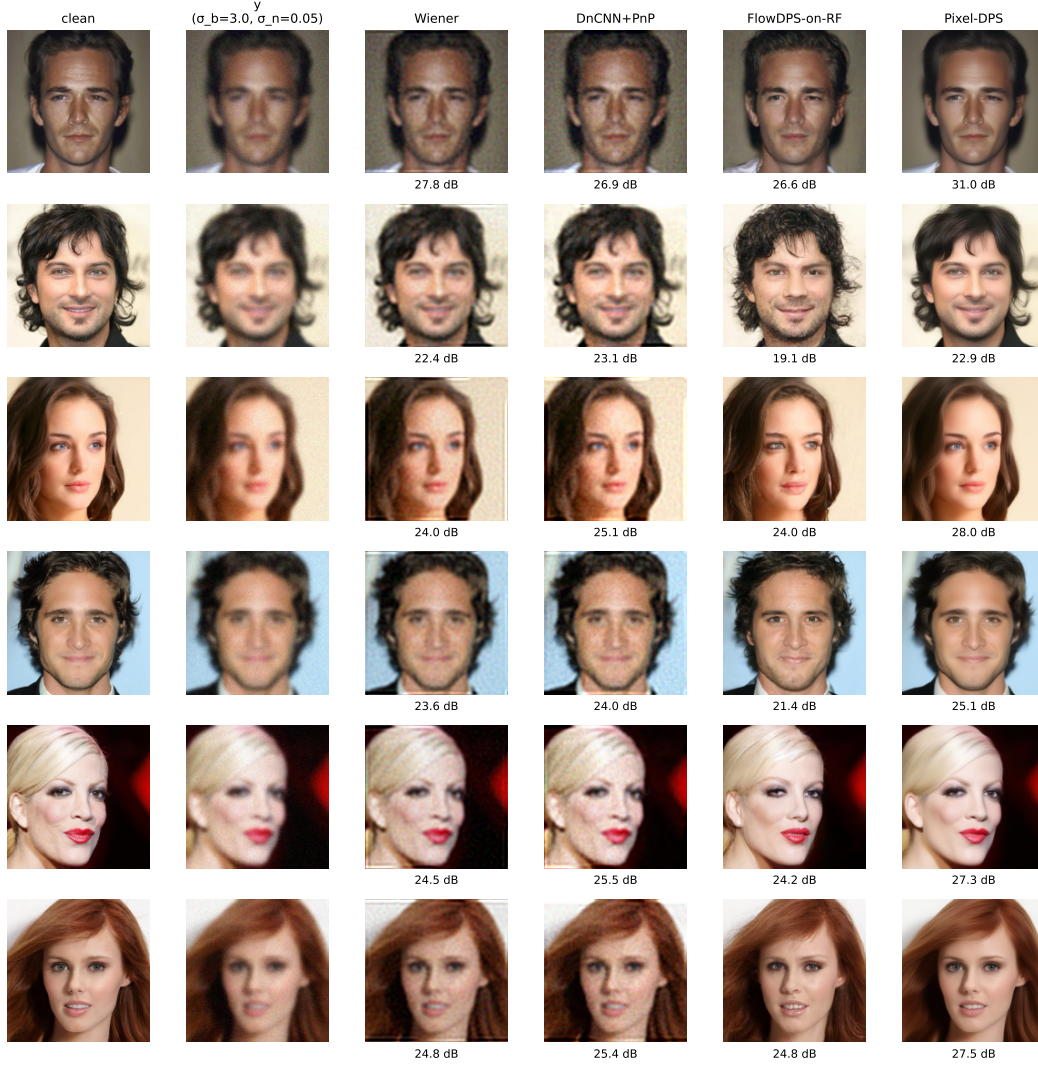


Figure 4: Qualitative comparison on six representative CelebA-HQ-256 test images at $\sigma_b=3.0$, $\sigma_n=0.05$, NFE=50. Columns: clean ground truth, degraded measurement, Wiener filter, DnCNN+PnP-ADMM, FlowDPS-on-RF, and Pixel-DPS. Numbers below each reconstruction are per-image PSNR (dB). Wiener and DnCNN+PnP are smooth but oversmooth identity-relevant detail; FlowDPS-on-RF sometimes drifts to a plausible-but-wrong face; Pixel-DPS most consistently recovers identity-preserving facial structure.

mismatched cell. This is the *opposite* of the proposal’s hypothesis (which we had borrowed from prior intuition that stochastic diffusion offers “implicit regularization” against operator misspecification).

Limitations. (i) 50 images per cell — error bars are reported as $\pm\sigma$ in the supplementary tables. (ii) Single dataset (CelebA-HQ-256 faces). (iii) Single forward-operator family per study (Gaussian / motion blur). (iv) Our DDRM reimplementation did not reach competitive PSNR in the available time (it collapsed to ~ 14 dB even at the easiest cell); we report DnCNN+PnP-ADMM as the analogous classical-prior reference instead. The parked DDRM code is preserved in the repository for future investigation. (v) The Liu 2023 RF checkpoint’s EMA weights are not loaded due to a shadow-params count mismatch in the released file; we use the non-EMA model state-dict, which may contribute ≤ 0.5 dB to the FlowDPS gap.

Future work. A clean DDRM implementation would close one gap in our baseline coverage. Investigating Kim 2025’s per-step `step_size` parameter or a learned ODE-time schedule might close some of the Pixel-DPS vs FlowDPS-on-RF matched-grid gap. Finally, repeating the comparison on non-face datasets (LSUN-Church, DIV2K) would test how much of the Pixel-DPS dominance is dataset-specific.

7 Conclusion

We presented a controlled, pixel-space comparison of Diffusion Posterior Sampling and a from-scratch reimplement of FlowDPS-style guidance on a Rectified Flow CelebA-HQ-256 prior. Pixel-DPS dominates the matched-Gaussian comparison by 1.4–4.2 dB PSNR across an 18-condition grid, but FlowDPS-on-RF is more robust to operator mismatch (−2.5 dB drop vs. Pixel-DPS’s −5 dB drop under severe motion-blur measurement mismatch). All four methods comfortably beat a Wiener-filter baseline; DnCNN+PnP-ADMM sits in the efficient middle of the Pareto frontier, beating FlowDPS-on-RF in 5/6 noisy cells while running $\sim 60\times$ faster than DPS. The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators: prefer DPS when the operator is calibrated; consider flow-based posterior sampling when the operator may be mis-specified.

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